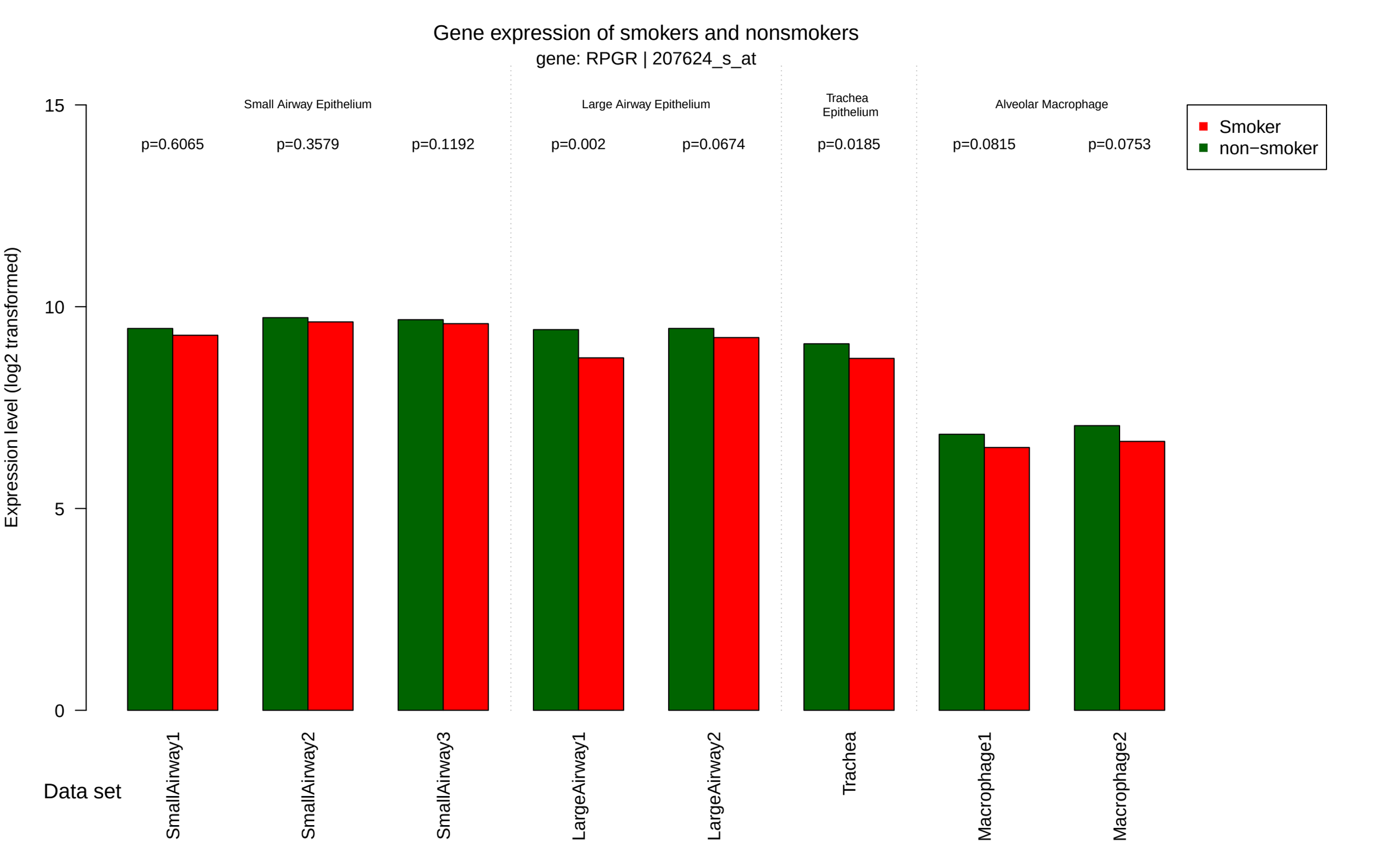
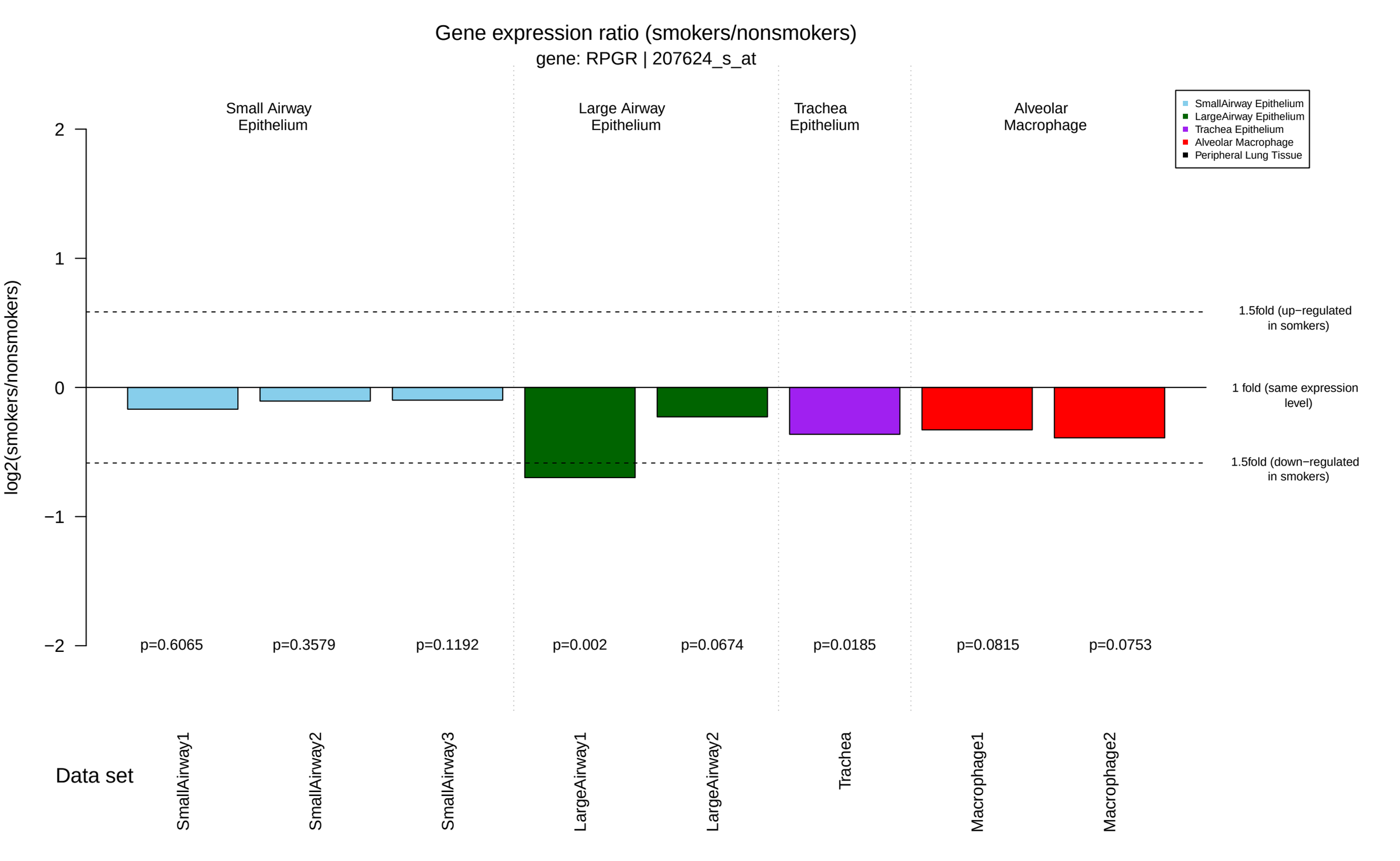
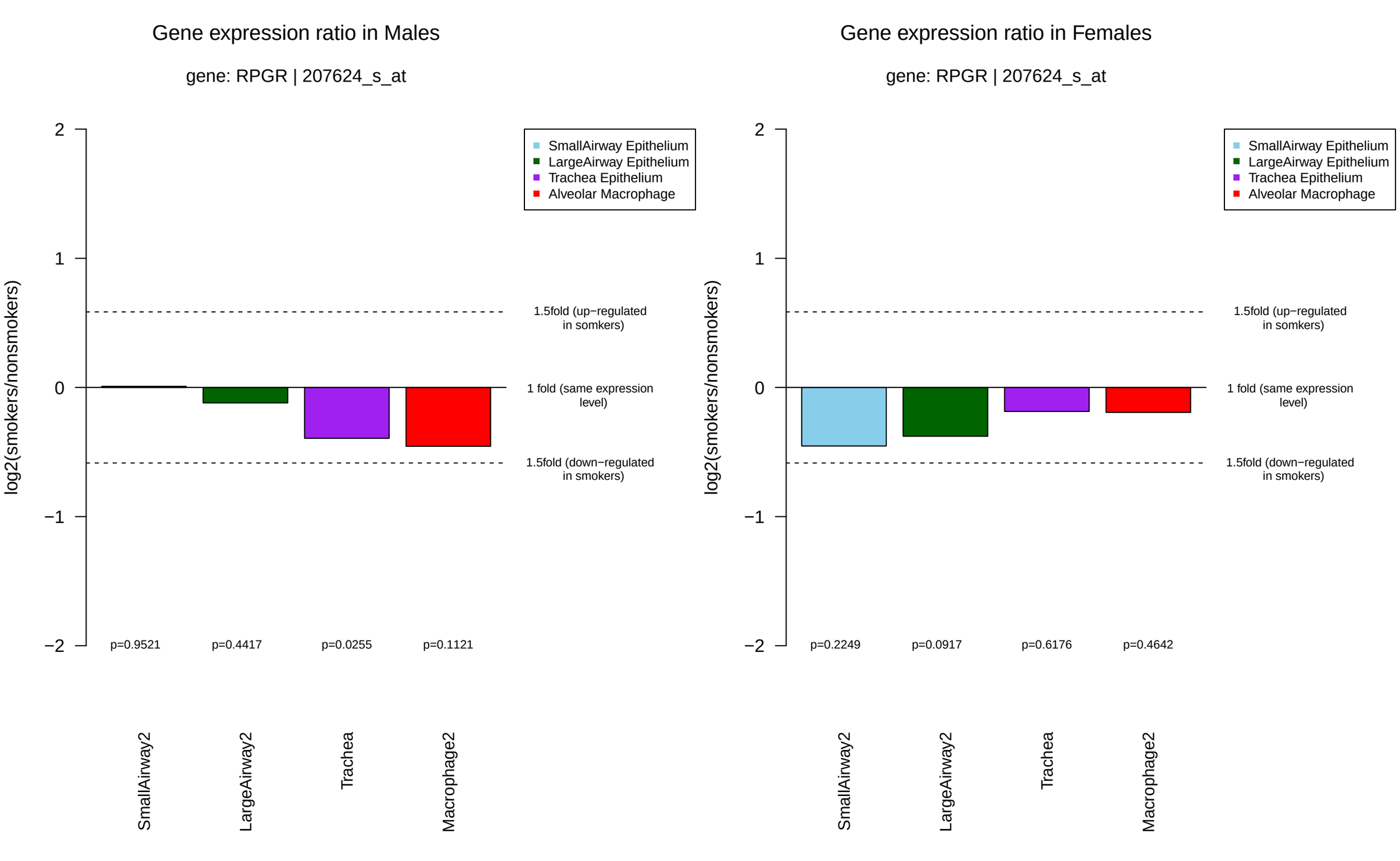
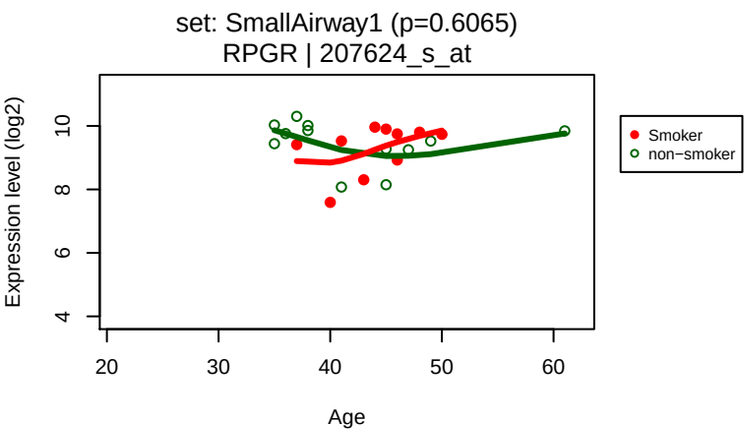
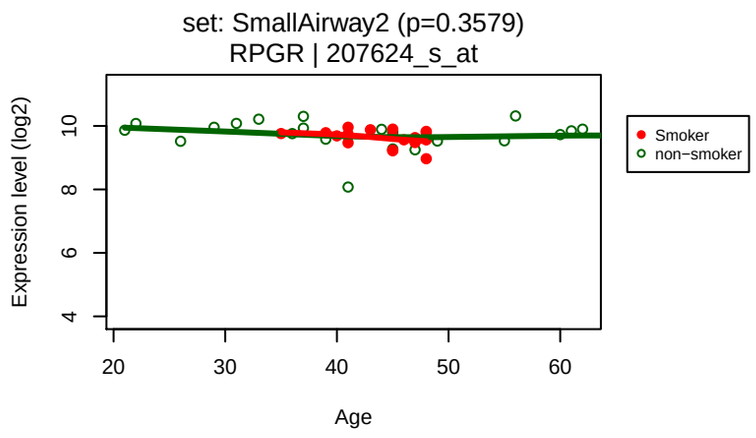
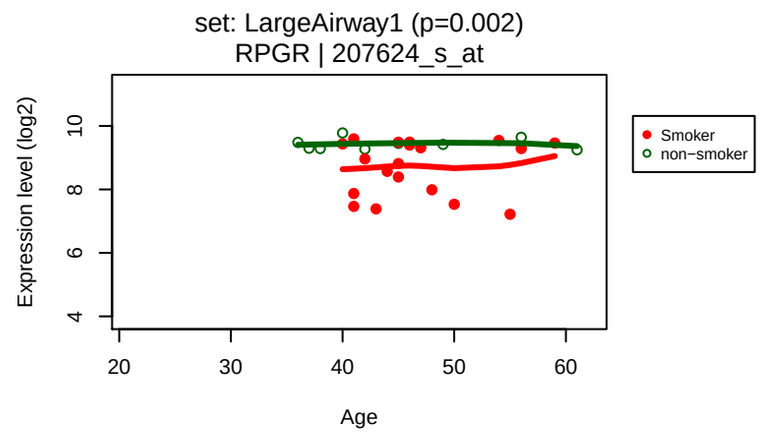

Gene expression of smokers and nonsmokers
gene: RPGR | 207624_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6065
p=0.3579
p=0.1192
p=0.002
p=0.0674
p=0.0185
p=0.0815
p=0.0753
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

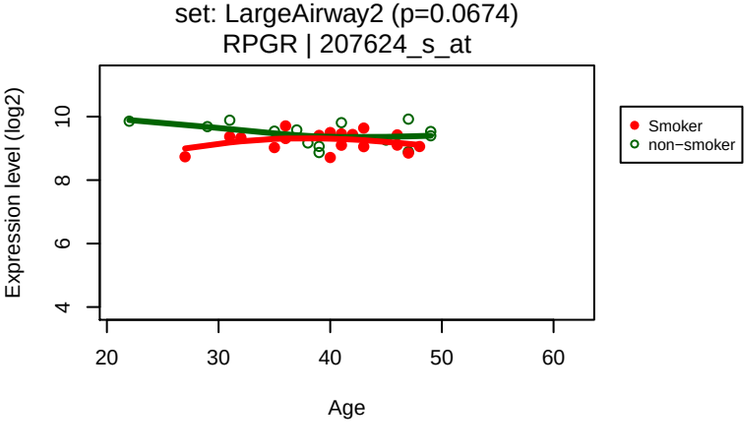

Gene expression ratio (smokers/nonsmokers)
gene: RPGR | 207624_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6065
p=0.3579
p=0.1192
p=0.002
p=0.0674
p=0.0185
p=0.0815
p=0.0753
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

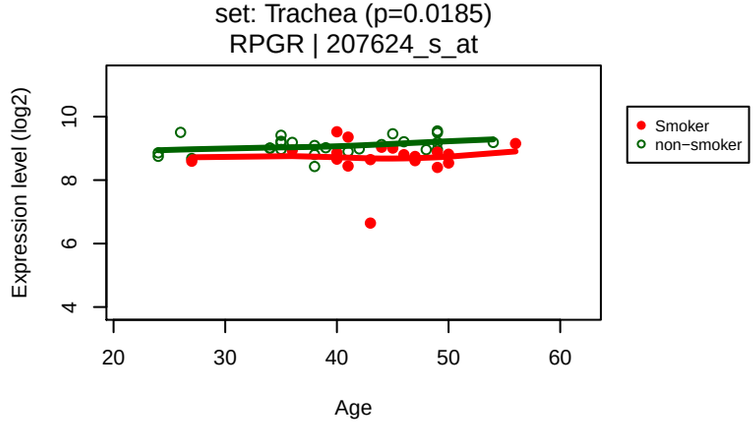

Gene expression ratio in Males
gene: RPGR | 207624_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9521
p=0.4417
p=0.0255
p=0.1121
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPGR | 207624_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2249
p=0.0917
p=0.6176
p=0.4642
SmallAirway2
LargeAirway2
Trachea
Macrophage2

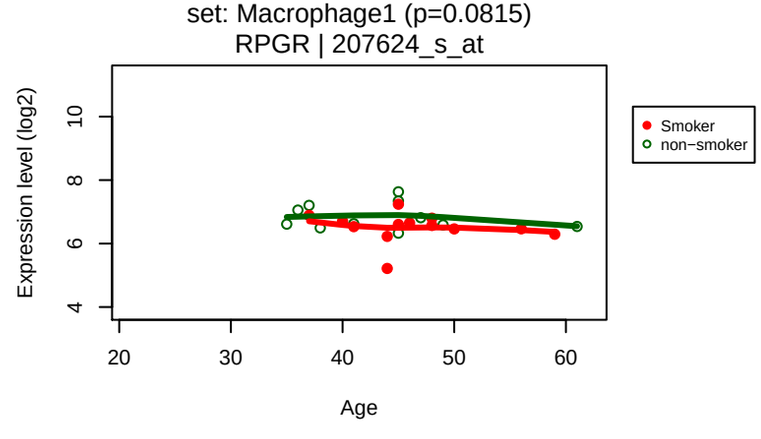

set: SmallAirway1 (p=0.6065)
RPGR | 207624_s_at
Expression level (log2)
Age
Smoker
non−smoker

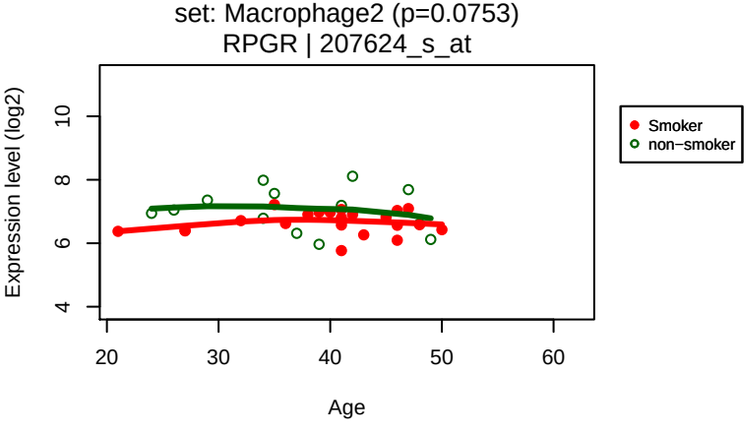

set: SmallAirway2 (p=0.3579)
RPGR | 207624_s_at
Expression level (log2)
Age
Smoker
non−smoker

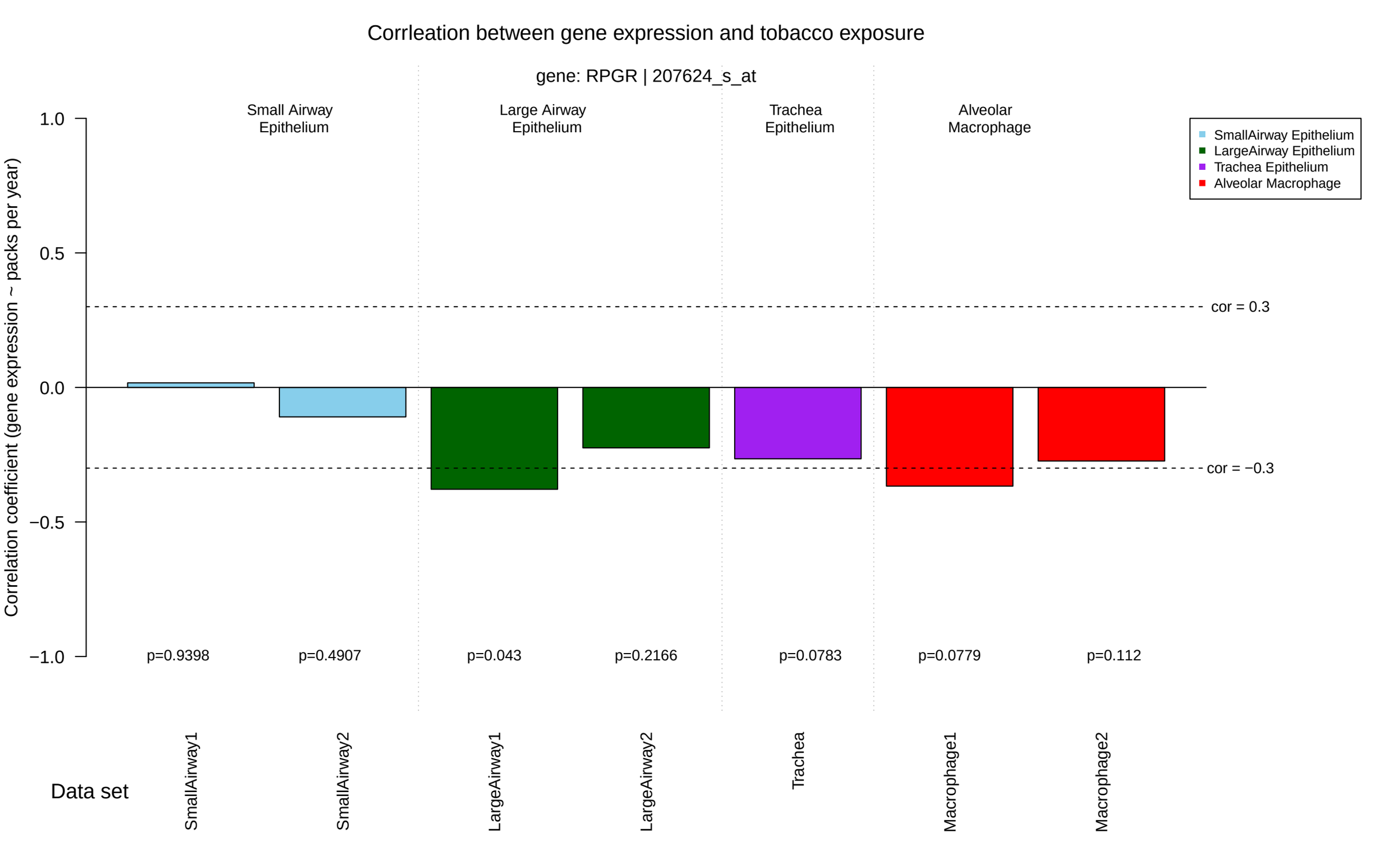

set: LargeAirway1 (p=0.002)
RPGR | 207624_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0674)
RPGR | 207624_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0185)
RPGR | 207624_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0815)
RPGR | 207624_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0753)
RPGR | 207624_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPGR | 207624_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9398
p=0.4907
p=0.043
p=0.2166
p=0.0783
p=0.0779
p=0.112
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2